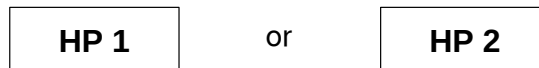


3. EMERGENCY AND MALFUNCTION PROCEDURES

3.1 CAUTION LIGHT INDICATIONS

CAUTION LIGHT INDICATIONS



Conditions/Indications

Disengage unit of self-centering circuit has failed

- **HP** caution light (affected computer) on

Procedure

1. Release pushbutton – Press

- NOTE**
- Immediately disengage AFCS by pressing either cyclic-stick release pushbutton. If not, an abrupt acceleration along the pitch axis may occur. Pressing the release pushbutton will cause the HP1, HP2 and YAW switches on the control panel to spring to the OFF position.
 - After an automatic or manual disengagement, however, the HELIPILOTS may be reengaged with the helicopter in a straight and level flight attitude. If, after reengagement, neither **HP** caution light comes on the mission can be continued normally.

4. NORMAL PROCEDURES

4.1 ★ AFCS CHECKS

- | | |
|------------------------------------|---|
| INV 1/INV 2 switches | – ON; INV 1/INV 2 caution lights off |
| DG switch | – ON |
| HP1/HP2 caution lights | – On |
| Instrument/indicator warning flags | – Out of view |
| ATT/SAS switch | – ATT |
| HP1 and YAW switches | – On; HP1 caution light off |
| API needles (PITCH, ROLL and YAW) | – Centered |

NOTE If API needles are not centered, press maneuver switch and check that needles center.

- | | |
|------------|--|
| HP1 switch | – OFF; YAW switch springs to OFF and HP1/HP2 caution lights come on |
|------------|--|

- | | |
|------------------------------|------------------------------------|
| HP2 switch | – On; HP2 caution light off |
| HP1/HP2 switch (API panel) | – HP2 and hold |
| API needles (PITCH and ROLL) | – Centered |

NOTE If API needles are not centered, press maneuver switch and check that needles center.

- | | |
|----------------------|---|
| HP1 and YAW switches | – On; HP1 and HP2 caution lights go off |
|----------------------|---|

★ To be checked each flight before takeoff.

4.2 PRE-TAKEOFF CHECK

- | | |
|----------------|-------|
| ATT/SAS switch | – SAS |
| YAW switch | – OFF |

NOTE YAW channel should be disengaged during takeoff and landing to reduce directional pedal control forces.

4.3 AFCS IN-FLIGHT OPERATION

4.3.1 Attitude Hold (ATT) Mode

- | | |
|---------------------|----------------------|
| YAW switch | – On |
| ATT/SAS switch | – ATT |
| Maneuver pushbutton | – Press, as required |

The maneuver switch located on the cyclic stick grip allows the pilot to manually take over control when the AFCS is engaged and a course or attitude change is desired. When the automatically controlled flight is to be continued, release the maneuver switch and trim the cyclic stick. Check that API needles are centered. If not, press maneuver switch and re-trim stick position if necessary.

NOTE The maneuver switch is also coupled with the cyclic trim so that the function of the maneuver switch is operative as long as the cyclic trim is activated. Therefore, it is advisable to apply cyclic control changes using trim. However, a readjustment of the actuator position may be necessary.

4.3.2 Stability Augmentation System (SAS) Mode

- | | |
|---------------------|----------------------|
| ATT/SAS switch | – ATT |
| Maneuver pushbutton | – Press, as required |

The centering of the actuators is possible during flight maneuvers by pressing the maneuver pushbutton. Maneuver pushbutton function is coupled with the cyclic trim system and

remains on for a further 3.5 seconds after releasing the four-way trim switch. However, a readjustment of the actuator position may be necessary. Therefore, check that API needles are centered. If not, press maneuver pushbutton and retrim stick position if necessary.

Automatic Actuator Centering:

When operating in the SAS mode, especially in turns with power reduction, the pitch actuators may run to their end of authority. The actuator centering circuit (actuator detector) will then initiate the centering of the actuators and bring them back to center within 3.5 seconds.

NOTE If the automatic-centering circuit fails, the respective AFCS computer will automatically disengage.

4.3.3 YAW SAS (YAW) Mode

The YAW SAS mode can be selected only when HP 1 and YAW channel are engaged. Indication that YAW SAS is selected is displayed by the position of toggle switches. Placing either the YAW or the HP 1 switches to OFF cancels the mode.

4.3.4 AFCS Trimming

Trim the AFCS and center the ROLL and PITCH API needles by placing the cyclic four-way trim switch in the direction the API needles point, e.g. when the ROLL API needle points towards the dot (to the right) push the four-way trim switch to the right until the needle centers. When the PITCH API needle points towards the dot (downwards) push the four-way trim switch forward, and so on.

When trimming the yaw axis, use the collective lever four-way YAW TRIM switch in conjunction with the ADI inclinometer. If the inclinometer ball deflects to the left, push the YAW TRIM switch to the left to center the ball; if the ball deflects to the right, push the switch to the right.

NOTE Yaw trimming is possible only when the YAW channel is engaged. Changing the display of the ROLL and PITCH APIs from pitch/roll actuators 1 to pitch/roll actuators 2 is possible with the spring-loaded HP1/HP2 switch on the API panel.

4.4 PRE-LANDING CHECK

ATT/SAS switch	– SAS
YAW switch	– OFF

NOTE YAW channel should be disengaged during takeoff and landing to reduce directional pedal control forces.